OCR
Oxford Cambridge and RSA

Monday 19 October 2020 – Afternoon

A Level Mathematics B (MEI)

H640/03 Pure Mathematics and Comprehension

Printed Answer Booklet

Time allowed: 2 hours

8168072005

You must have:
- Question Paper H640/03 (inside this document)
- the Insert (inside this document)
- a scientific or graphical calculator

H640003

Please write clearly in black ink. **Do not write in the barcodes.**

Centre number | | | | | | Candidate number | | | | |

First name(s) _____

Last name _____

INSTRUCTIONS
- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined pages at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give your final answers to a degree of accuracy that is appropriate to the context.

INFORMATION
- This document has **16** pages.

ADVICE
- Read each question carefully before you start your answer.

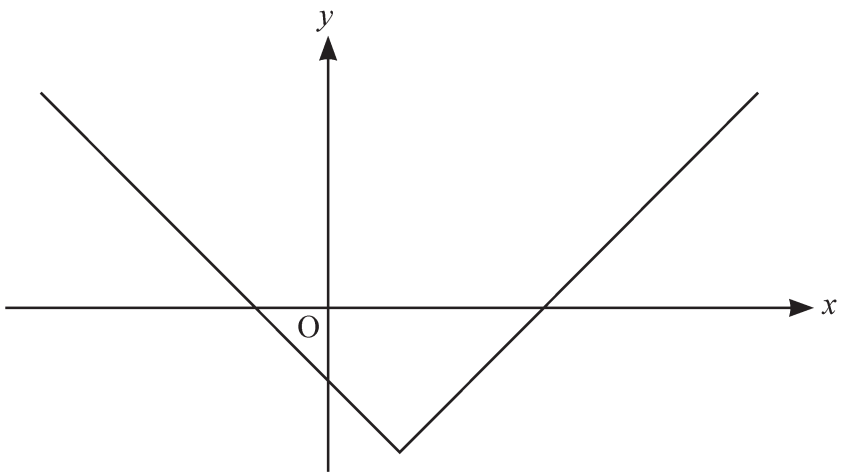
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Section A (60 marks)

1

2



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3

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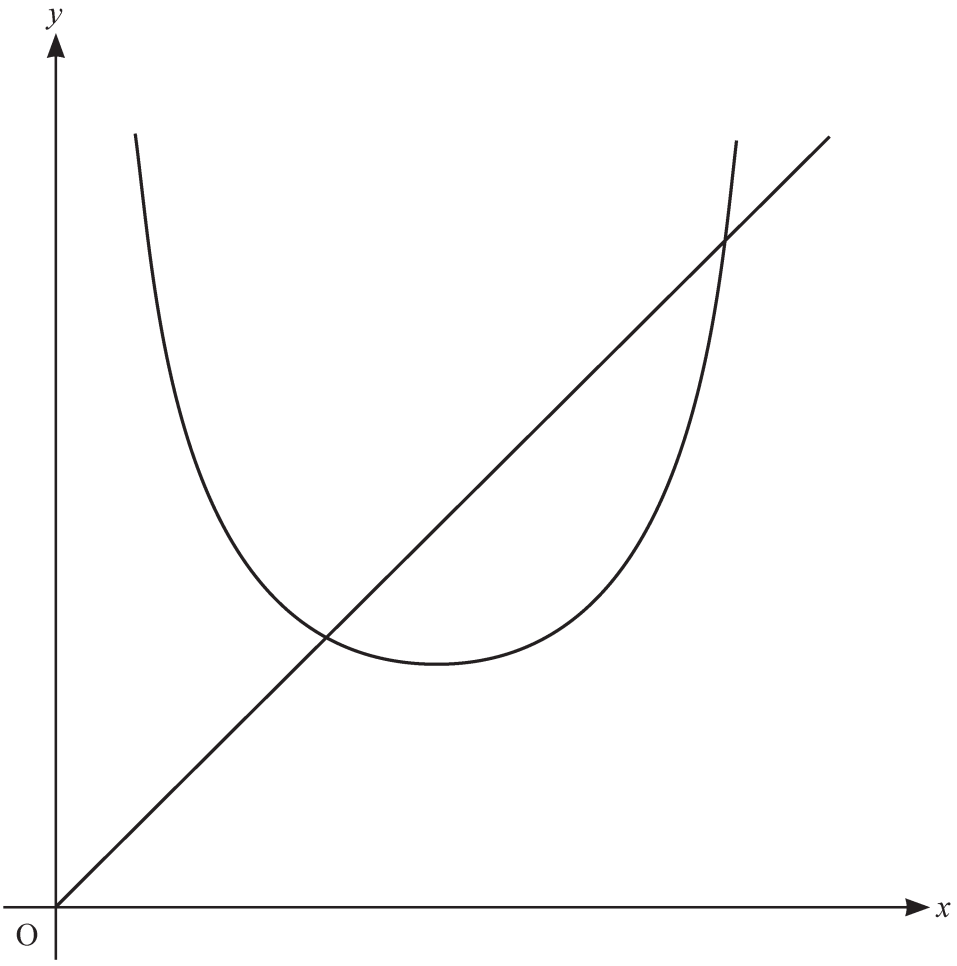
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5(a)(i)	
5(a)(ii)	
5(b)	
5(c)(i)	
5(c)(ii)	

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5(d)

5(e)



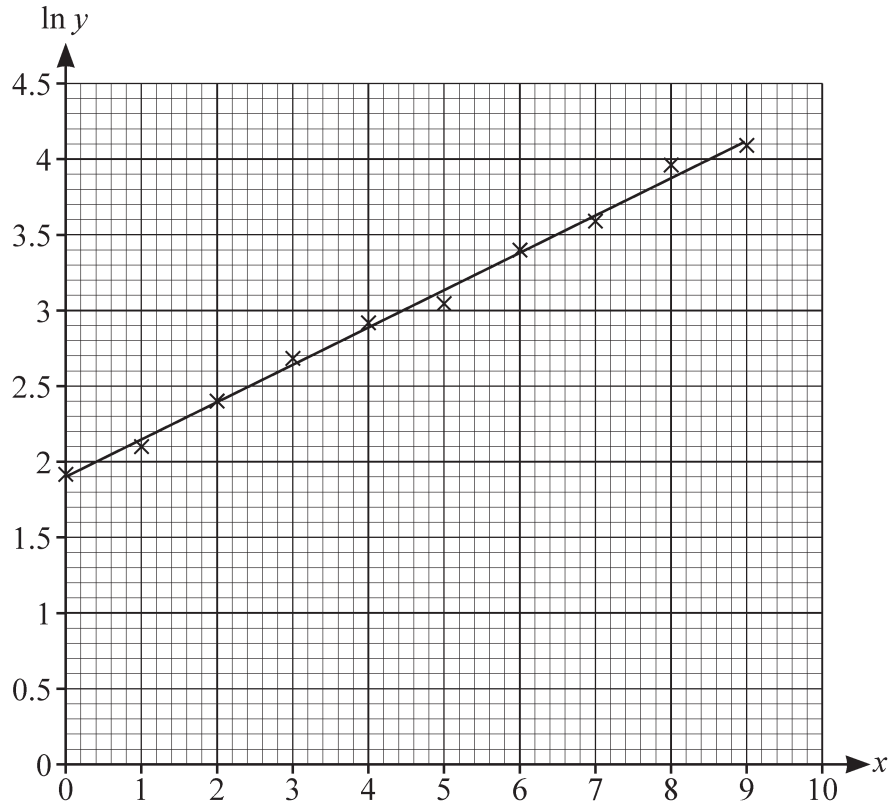
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6(a)(i)

6(a)(ii)

6(b)

6(c)



(answer space continued on next page)

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6(c)	**(continued)**
6(d)	
6(e)	

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7(a)

7(b)

(answer space continued on next page)

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7(b)	(continued)

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8(a)(i)

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8(a)(ii)

8(b)

(answer space continued on next page)

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8(b)	(continued)

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Section B (15 marks)

The questions in this section refer to the article on the Insert. You should read the article before attempting the questions.

9 **(a)** Show that if $a = 1$ and $b > 1$ then $a^b < b^a$. **[2]**

(b) Find integer values of a and b with $b > a > 1$ and a^b not greater than b^a (a counter example to the conjecture given in lines 7–8). **[1]**

9(a)	
9(b)	

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10 In this question you must show detailed reasoning.

Show that $\int_{\mathrm{e}}^{\pi} \frac{1}{x}\,\mathrm{d}x = \ln\pi - 1$ as given in line 37. **[2]**

10	

11 Show that e^x is an increasing function for all values of x, as stated in line 39. **[2]**

11	

12 (a) Show that the only stationary point on the curve $y = \frac{\ln x}{x}$ occurs where $x = \mathrm{e}$, as given in line 45. **[3]**

(b) Show that the stationary point is a maximum. **[3]**

(c) It follows from part **(b)** that, for any positive number a with $a \neq \mathrm{e}$,

$$\frac{\ln \mathrm{e}}{\mathrm{e}} > \frac{\ln a}{a}.$$

Use this fact to show that $\mathrm{e}^a > a^{\mathrm{e}}$. **[2]**

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12(a)

12(b)

12(c)

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ADDITIONAL ANSWER SPACE

If additional space is required, you should use the following lined page(s). The question number(s) must be clearly shown in the margin(s).

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